Saee Saadat

Software Engineer

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About Me

Mid-Level Software Engineer focusing on building reliable backend services, mostly using Python and Java. System Design, Database Design, and Secure Software Engineering are my technical strong suits. A proactive team member who loves brainstorming ways to improve both the product and internal workflows, constantly learning and pushing to grow, with the goal of bringing Senior-level technical depth and mentorship to everything I do.

Work Experience

PSI Software 07/2025 – Present
Working Student, Software Engineer *Düsseldorf, Germany*

- Led the technical development of the company's **AI Agent & Chatbot** initiative, architecting a multi-agent hierarchy and agentic workflows on Google ADK, and developing multiple tools and MCP servers to extend agent capabilities.
- Worked with Java, Maven, Quarkus, PostgreSQL, Python, Google ADK, FastMCP, K8s, AngularJS, Google Cloud and Keycloak on an Enterprise multi-modular Java-based MES for the metals industry.
- Created multiple temporal resampling algorithms for time-series data, resulting in up to **70% faster** chart rendering.
- Drove adoption of agentic technologies across the team by onboarding and mentoring developers and collaborating with domain experts to build specialised AI agents.
- Created a course for colleagues on how to build agent skills and co-created a skill registry for sharing them.
- Collaborated with security experts to resolve security issues in the codebase and in the team's everyday tools.

AYOT 07/2021 – 09/2024
Founder, Software Engineer *Sharif Innovation Park, Tehran*

- Founded an IoT & smart home startup; recruited and led a team of **10 professional engineers**.
- Co-designed the MVP and the final product's architecture. Tech stack: Python, FastAPI, PostgreSQL, Redis, K8s, Flutter, MQTT and Matter, deployed on VMs managed by a cloud provider.
- Applied TDD (100% test coverage), lean agile methodology, and full project management: backlog, Kanban board, GitHub Actions, and complete software documentation.
- Defined use-cases and user stories, performed risk analysis, and managed the full product backlog.

Digikala 04/2022 – 07/2023
Software Engineer *Tehran, Iran*

- Worked on customer facing and vendor facing components of Iran's largest e-commerce platform (legacy codebase, 1M+ lines).
- Maintained high-traffic backend serving **50,000+ daily active users**. Tech stack: PHP, Symfony, TypeScript, MySQL, CQRS, Mono-repo, Jenkins, Docker, Redis, Kafka.
- Started the redesign process; co-designed the new Microservice architecture using Golang, Elasticsearch and MongoDB.
- Re-implemented the OSM-based delivery estimation service using ALT instead of static caching, reducing cancelled orders by **10%** on adverse weather days.
- Implemented a parallel delivery assignment and tracking service, resulting in **15% faster** delivery time.

Ostad Kar 07/2021 – 06/2022
Software Engineer *Tehran, Iran*

- Maintained and developed features for the core backend components. Tech stack: Python, Flask, Docker, PostgreSQL, Cassandra, Redis, RabbitMQ.
- Created an automated report system that integrates with 3rd party providers.
- Maintained the CI/CD pipeline and developed skills across the DevOps and deployment side of software.

Farazpardazan 02/2020 – 01/2021
iOS App Developer *Tehran, Iran*

- Joined the Eghtesad Novin Bank mobile banking iOS team after 3 months of training (Swift, RxSwift, Alamofire, fastlane).
- Took full ownership of the Razi Insurance iOS app with no senior developer to supervise, during my 4th semester.

Education

University of Bonn 10/2024 – 09/2027 (expected)
M.Sc. in Computer Science (GPA: 1.6) *Bonn, Germany*

Sharif University of Technology 09/2018 – 02/2024
B.Sc. in Computer Engineering (GPA: 1.5) *Tehran, Iran*

- Top engineering university in Iran. Completed two co-op semesters of full-time work alongside studies and additional entrepreneurial coursework.

Projects

Independent Backend System | *Java, Quarkus, GCP, Terraform, PostgreSQL, Redis, S3* 2025 – 2026

- Sole architect and tech lead on a production-grade Java/Quarkus backend with Domain-Driven Design, translating a non-technical stakeholder's use cases into user stories through full requirement engineering.
- Built custom authentication and authorisation from scratch, full REST API layer, domain layer, async/scheduled task management, Redis caching, and S3-based media storage.
- Deployed and managed dev and production environments on GCP with all infrastructure as code using Terraform.
- Acted as tech lead, coordinating a designer and front-end developer and owning the full technical direction.

Scientific Survey Platform | *FastAPI, Python, Yjs, WebSockets* 2024 – Present

- Active contributor to a real-time scientific survey platform for the Behavioral Security Research Group at the University of Bonn.
- Integrated Yjs for live multi-user collaboration, enabling conflict-free simultaneous editing by multiple researchers.

Personal IoT Platform | *MQTT, Python, Arduino, Home Assistant* 2023 – Present

- Built a personal MQTT-based IoT platform connecting custom Arduino gadgets over Wi-Fi, with a Python backend managing device state and events.
- Implemented asymmetric encryption and device authentication from scratch to secure all device communication.

Certifications

Engineer AI Agents with Agent Development Kit (ADK), Google Cloud, May 2026

Technical Skills

Languages: Python, Java, Kotlin, Go, PHP, TypeScript, Swift
Frameworks: FastAPI, Quarkus, Spring Boot, Symfony, Flask, Gin, AngularJS, Flutter
Cloud & DevOps: Docker, K8s, GitHub Actions, GitLab CI/CD, Jenkins, Google Cloud, Terraform
Databases: PostgreSQL, MySQL, Redis, MongoDB, CassandraDB, Elasticsearch, S3, pgvector
Messaging & Protocols: Kafka, RabbitMQ, REST-API, gRPC, MQTT, Matter, WebSockets, MCP
AI & Agents: Google ADK, FastMCP, LangChain, multi-agent systems, agentic workflow design, RAG, Claude Code, Gemini CLI
Security: Secure software design, zero-trust principles, encryption, Keycloak
Project Management: Scrum/Agile, TDD/BDD, Kanban, Jira, GitHub Projects

Soft Skills

Leadership & Mentoring: I led a team of 10 engineers as a startup founder and onboarded and mentored developers at PSI Software.
Ownership & Initiative: I proactively identify problems and drive solutions end-to-end, from spotting issues in production systems to proposing and shipping the fix.
Cross-functional Collaboration: I work effectively with domain experts, designers and non-technical stakeholders, and I translate use cases into engineering requirements.
Communication: I have experience in technical onboarding, requirement engineering, risk analysis and writing complete software documentation.

Language Proficiency

English: C2 (Fluent) **German**: A2 (actively improving) **Persian**: Native

Interests & Hobbies

Padel, IoT passion projects with Arduino, Gym, Soccer, Table Tennis, Cooking, Tech news